

OPERATING SYSTEM

Complete Notes · MCQ + Written · BB AD + All Banks · 2019–2026

Total Questions	BB AD Written	Topics	Prediction
21+ Questions	Banker's (2025) + Scheduling	CPU Scheduling · Banker's · Deadlock · Memory	CPU Scheduling 90% next exam

01 কী পড়বে — Priority + Topic Map

OS থেকে BB AD Written-এ সরাসরি ২টা major type আসে: (1) Banker's Algorithm, (2) CPU Scheduling। MCQ-তে আসে Process States, Page Fault, Deadlock concepts।

Topic	BB Priority	MCQ/Written	Next Exam Chance	Days
CPU Scheduling (FCFS/SJF/RR/Priority)	P1 — MUST	Written (Q4 slot)	90% — rotation after Banker's 2025	2 days
Banker's Algorithm	P1 — MUST	Written (Q4 slot)	70% — already came 2025	1.5 days
Process States & Transitions	P1	MCQ (every year)	95% MCQ	0.5 day
Deadlock (4 conditions + prevention)	P2	MCQ + Short Written	75%	1 day
Memory Management (Paging/Virtual)	P2	MCQ	60% MCQ	1 day
Semaphore / Critical Section	P3	MCQ / Other Bank Written	50%	0.5 day
Multiprogramming / SPOOLING / Interrupt	P3	MCQ	40%	0.5 day

BB NOTE: BB AD 2025-এ Banker's Algorithm Written-এ এসেছে। Rotation theory অনুযায়ী next exam-এ CPU Scheduling (FCFS/SJF/Round Robin) আসার probability ৯০%। কিন্তু Banker's এর পরীক্ষায় আবার Banker's আসতে পারে — তাই দুটোই master করতে হবে।

02 Process Basics — MCQ Foundation

Process কী?

Process = একটা Program যেটা currently executing হচ্ছে। Program = disk-এ stored code (passive)। Process = memory-তে loaded এবং চলছে (active)।

Process States — 5-State Model

State	বাংলায় মানে	কখন এই state-এ যায়
New	নতুন তৈরি হচ্ছে	Process সবে create হয়েছে, OS admit করেনি
Ready	CPU-র জন্য অপেক্ষা	CPU ছাড়া সব resource আছে। Scheduler-এর queue-এ
Running	CPU-তে execute হচ্ছে	Scheduler এই process-কে CPU দিয়েছে
Waiting (Blocked)	I/O-র জন্য অপেক্ষা	I/O operation দরকার (disk read, network, keyboard)
Terminated	শেষ	Process complete হয়েছে বা killed হয়েছে

BB NOTE: BB AD 2025 MCQ: "A process needs I/O operations, it switches to?" = Waiting (Blocked)। এই question প্রতি বছর MCQ-তে আসে। Running → Waiting: I/O request। Waiting → Ready: I/O complete। Running → Ready: Time slice expire (preemption)।

Key OS Concepts — MCQ Quick Reference

Concept	সংজ্ঞা	MCQ Answer
Multiprogramming	একসাথে multiple programs memory-তে। CPU idle থাকে না।	Execute more jobs in same time period
Multitasking	User-এর দৃষ্টিকোণ থেকে একসাথে multiple tasks। Time-sharing।	Better user experience with concurrent execution
Context Switch	CPU এক process থেকে আরেকটায় switch করা। State save+restore।	Overhead — Round Robin-এ সবচেয়ে বেশি
Spooling	I/O data একটা buffer-এ রাখা। Printer spooling সবচেয়ে common।	Simultaneous Peripheral Operations On-Line
Interrupt	Hardware/Software event যা CPU-কে current task pause করায়।	Hardware-generated signal to CPU
Kernel	OS-এর core part। Hardware directly manage করে।	Core of OS — most privileged mode

03 CPU Scheduling — Full Coverage (Next Exam P1)

CPU Scheduling নিয়ে BB AP 2023-এ MCQ এসেছে। Next BB AD-এ Written হিসেবে আসার সম্ভাবনা সবচেয়ে বেশি। এই section-এ সব algorithm, সব formula, এবং multiple solved problems আছে।

Key Formulas — এগুলো মুখস্থ করতেই হবে

AT = Arrival Time (কখন process আসে)

BT = Burst Time (CPU কত সময় দরকার)

CT = Completion Time (কখন শেষ হয়)

TAT = Turnaround Time = $CT - AT$ (কতক্ষণ সিস্টেমে ছিল)

WT = Waiting Time = $TAT - BT$ (কতক্ষণ CPU-র জন্য অপেক্ষা করেছে)

Average TAT = $(\text{Sum of all TAT}) \div \text{Number of processes}$

WT = $(\text{Sum of all WT}) \div \text{Number of processes}$

4 Scheduling Algorithms — Comparison Table

Algorithm	কীভাবে কাজ করে	Preemptive?	সুবিধা	অসুবিধা	BB Note
FCFS	Arrival order-এ execute	No	Simple, no starvation	Convoy effect — short jobs wait for long	Most basic — always in exam
SJF	Shortest Burst Time আগে	No (SRTF = Yes)	Minimum average WT (optimal)	Starvation of long jobs	BB MCQ: causes starvation
Priority	Highest priority (lowest number) আগে	Yes/No	Important jobs আগে	Starvation of low-priority jobs	BB AP 2023 Q — exactly this
Round Robin	Fixed time quantum করে each process-কে দেওয়া হয়	Yes	Fair, no starvation, time-sharing	High context switch overhead, quantum size critical	Best for time-sharing systems

3A — FCFS (First Come First Served) — Solved

QUESTION: Given: P1(AT=0,BT=24), P2(AT=1,BT=3), P3(AT=2,BT=3). FCFS scheduling — Draw Gantt Chart. Find Average TAT and WT.

কীভাবে ভাববেন: FCFS-এ arrival order অনুসরণ করো। AT=0 থেকে শুরু। কোনো preemption নেই — একটা শেষ না হলে অন্যটা শুরু হবে না।

SOLVED — FCFS Scheduling

Execution order: P1 → P2 → P3 (arrival order)

Gantt Chart: | P1 | P2 | P3 |

0 24 27 30

Completion Times (CT):

P1: 24 P2: 27 P3: 30

Turnaround Time (TAT = CT - AT):

P1: 24-0 = 24

P2: 27-1 = 26

P3: 30-2 = 28

Waiting Time (WT = TAT - BT):

P1: 24-24 = 0

P2: 26-3 = 23

P3: 28-3 = 25

Average TAT = $(24+26+28) \div 3 = 78 \div 3 = 26$ ms

Average WT = $(0+23+25) \div 3 = 48 \div 3 = 16$ ms

Note: "Convoy Effect" — P2 and P3 (short jobs) wait long time for P1 (long job). SJF would be much better here.

3B — SJF (Shortest Job First) Non-Preemptive — Solved

QUESTION: Same processes: P1(AT=0,BT=6), P2(AT=1,BT=8), P3(AT=2,BT=7), P4(AT=3,BT=3). Non-Preemptive SJF. Find Average WT.

কীভাবে ভাববেন: Non-Preemptive SJF: যখন CPU free হয়, সেই মুহুর্তে যত processes arrived আছে তাদের মধ্যে সবচেয়ে short BT-এর job পরবর্তী।

SOLVED — Non-Preemptive SJF

Time 0: Only P1 arrived → run P1 (BT=6). Runs 0–6. Time 6: CPU free.

Arrived: P2(8), P3(7), P4(3). Shortest = P4(3) → run P4. Runs 6–9. Time 9: CPU free.

Arrived: P2(8), P3(7). Shortest = P3(7) → run P3. Runs 9–16. Time 16: CPU free.

Only P2 left → run P2. Runs 16–24.

Gantt Chart: | P1 | P4 | P3 | P2 |
0 6 9 16 24

CT: P1=6, P2=24, P3=16, P4=9

TAT = CT – AT: P1: 6–0=6 P2: 24–1=23 P3: 16–2=14 P4: 9–3=6

Average TAT = $(6+23+14+6) \div 4 = 49 \div 4 = 12.25$ ms

WT = TAT – BT: P1: 6–6=0 P2: 23–8=15 P3: 14–7=7 P4: 6–3=3

Average WT = $(0+15+7+3) \div 4 = 25 \div 4 = 6.25$ ms

3C — Priority Scheduling — BB AP 2023 Exact

QUESTION: BB AP 2023: Priority Scheduling (Non-Preemptive). P1(BT=10,P=3), P2(BT=1,P=1), P3(BT=2,P=4), P4(BT=1,P=5), P5(BT=5,P=2). All AT=0. Lower number = higher priority. Average TAT?

কীভাবে ভাববেন: Priority number ছোট = priority বেশি। No arrival time মানে সব একসাথে শুরু। তাই সরাসরি priority order অনুযায়ী execute করো।

SOLVED — BB AP 2023 — Priority Scheduling (EXACT ANSWER)

Execution Order (sort by priority ascending = lower number first):

P2(P=1) → P5(P=2) → P1(P=3) → P3(P=4) → P4(P=5)

Gantt Chart: | P2 | P5 | P1 | P3 | P4 |
0 1 6 16 18 19

Completion Times:

P1: 16 P2: 1 P3: 18 P4: 19 P5: 6

TAT = CT – AT (AT=0 for all):

P1: 16 P2: 1 P3: 18 P4: 19 P5: 6

Average TAT = $(16+1+18+19+6) \div 5 = 60 \div 5 = 12$

Answer: (c) 12 ← BB AP 2023-এর exact answer!

3D — Round Robin — Solved

QUESTION: Round Robin: P1(BT=20), P2(BT=3), P3(BT=3). Time Quantum = 2. All AT=0. Find TAT and WT.

কীভাবে ভাববেন: Round Robin-এ প্রতিটা process quantum time পায়। Remaining time track করো। $BT \leq \text{quantum}$ হলে সেটা শেষ হয়ে যায়।

SOLVED — Round Robin (TQ=2)

Remaining BT:

P1=20, P2=3, P3=3

Cycle 1: P1 gets 2 → remaining P1=18 | Time: 0–2

Cycle 2: P2 gets 2 → remaining P2=1 | Time: 2–4

Cycle 3: P3 gets 2 → remaining P3=1 | Time: 4–6

Cycle 4: P1 gets 2 → remaining P1=16 | Time: 6–8

Cycle 5: P2 gets 1 → P2 done! CT=9 | Time: 8–9

Cycle 6: P3 gets 1 → P3 done! CT=10 | Time: 9–10

Cycle 7-13: P1 gets remaining 16 | Time: 10–26. CT=26

Gantt: |P1|P2|P3|P1|P2|P3|P1|P1|P1|P1|P1|P1|P1|

0 2 4 6 8 9 10 12 14 16 18 20 22 24 26

TAT:

$P1=26-0=26$, $P2=9-0=9$, $P3=10-0=10$

Avg TAT = $(26+9+10) \div 3 = 45 \div 3 = 15$ ms

WT: $P1=26-20=6$, $P2=9-3=6$, $P3=10-3=7$

Avg WT = $(6+6+7) \div 3 = 6.33$ ms

TIP: Exam-এ Gantt Chart আঁকলেই অর্ধেক marks পাওয়া যায় — এমনকি calculation ভুল হলেও। সবসময় Gantt Chart দিয়ে শুরু করো।

04 Banker's Algorithm — BB AD 2025 Written Q4

Banker's Algorithm হলো Deadlock Avoidance technique। OS একটা "banker" (ব্যংকার) এর মতো কাজ করে — resource দেওয়ার আগে দেখে system safe state-এ থাকবে কিনা।

Key Concepts + Matrices

Matrix/Term	মানে কী	কীভাবে পাবে
Allocation	প্রতিটা process কতটা resource currently hold করছে	Given in problem
Max	প্রতিটা process সর্বোচ্চ কতটা resource চাইতে পারে	Given in problem
Need	প্রতিটা process আরও কতটা resource চাইতে পারে	$\text{Need} = \text{Max} - \text{Allocation}$
Available	বর্তমানে কতটা resource free আছে	Given (or $\text{Total} - \text{Sum of Allocation}$)
Safe State	একটা execution order আছে যেখানে সব process complete হতে পারে	Safety Algorithm দিয়ে check করো
Safe Sequence	যে order-এ processes execute হলে deadlock হবে না	Safety Algorithm-এর output

Safety Algorithm — 3 Steps

Step 1: $\text{Work} = \text{Available}$ (initially)

$\text{Finish}[i] = \text{false}$ for all i

Step 2: Find an i such that:

$\text{Finish}[i] = \text{false}$ AND $\text{Need}[i] \leq \text{Work}$

If found $\rightarrow \text{Work} = \text{Work} + \text{Allocation}[i]$, $\text{Finish}[i] = \text{true} \rightarrow \text{repeat}$

Step 2 If not found $\rightarrow$ go to Step 3

Step 3: If $\text{Finish}[i] = \text{true}$ for all $i \rightarrow \text{SAFE STATE} \checkmark$

If any $\text{Finish}[i] = \text{false} \rightarrow \text{UNSAFE (potential deadlock)} \times$

BB AD 2025 Written Q4 — Exact Banker's Problem**QUESTION:** BB AD 2025 Written Q4: Banker's Algorithm.

5 processes P0-P4.

3 resource types A(10), B(5), C(7).

Allocation: P0(0,1,0) P1(2,0,0) P2(3,0,2) P3(2,1,1) P4(0,0,2)

Max: P0(7,5,3) P1(3,2,2) P2(9,0,2) P3(2,2,2) P4(4,3,3)

Available: A=3, B=3, C=2

(a) Find Need matrix. (b) Is system in safe state? If yes, find safe sequence.

কীভাবে ভাববেন: এই question-এ প্রথমেই Need matrix বের করো: $\text{Need} = \text{Max} - \text{Allocation}$ তারপর Safety Algorithm চালাও। $\text{Work} = \text{Available}$ থেকে শুরু করে — $\text{Need} \leq \text{Work}$ এমন process খোঁজো।**SOLVED — BB AD 2025 Q4 — Banker's Algorithm Complete Solution**(a) Need Matrix ($\text{Need} = \text{Max} - \text{Allocation}$):

A B C

P0: $7-0 \ 5-1 \ 3-0 = 7 \ 4 \ 3$ P1: $3-2 \ 2-0 \ 2-0 = 1 \ 2 \ 2$ P2: $9-3 \ 0-0 \ 2-2 = 6 \ 0 \ 0$ P3: $2-2 \ 2-1 \ 2-1 = 0 \ 1 \ 1$ P4: $4-0 \ 3-0 \ 3-2 = 4 \ 3 \ 1$

(b) Safety Algorithm:

Initial: $\text{Work} = [3, 3, 2]$ $\text{Finish} = [F, F, F, F, F]$ Step 1: Check each process — $\text{Need} \leq \text{Work}$?P0: $\text{Need}=[7,4,3] \leq [3,3,2]$? NO ($7>3$)P1: $\text{Need}=[1,2,2] \leq [3,3,2]$? YES! ($1 \leq 3, 2 \leq 3, 2 \leq 2$) → Run P1. $\text{Work} = \text{Work} + \text{Alloc}[P1] = [3,3,2] + [2,0,0] = [5,3,2]$. $\text{Finish}[P1]=T$ Step 2: $\text{Work} = [5,3,2]$ P0: $\text{Need}=[7,4,3] \leq [5,3,2]$? NO ($7>5, 4>3$)P2: $\text{Need}=[6,0,0] \leq [5,3,2]$? NO ($6>5$)P3: $\text{Need}=[0,1,1] \leq [5,3,2]$? YES! → Run P3. $\text{Work} = [5,3,2] + [2,1,1] = [7,4,3]$. $\text{Finish}[P3]=T$ Step 3: $\text{Work} = [7,4,3]$ P0: $\text{Need}=[7,4,3] \leq [7,4,3]$? YES! → Run P0. $\text{Work} = [7,4,3] + [0,1,0] = [7,5,3]$. $\text{Finish}[P0]=T$

Step 4:

 $\text{Work} = [7,5,3]$ P2: $\text{Need}=[6,0,0] \leq [7,5,3]$? YES! → Run P2.

Work = $[7,5,3] + [3,0,2] = [10,5,5]$.

Finish[P2]=T

Step 5:

Work = $[10,5,5]$

P4: Need= $[4,3,1] \leq [10,5,5]$? YES! → Run P4.

Work = $[10,5,5] + [0,0,2] = [10,5,7]$.

Finish[P4]=T All Finish[i] = True → SAFE STATE!

Safe Sequence: P1 → P3 → P0 → P2 → P4

BB NOTE: BB AD 2025-এর exact question থেকে এই problem। Note matrix calculation-এ ভুল সবচেয়ে বেশি হয়। Need = Max - Allocation। একটু মাথা ঠান্ডা রেখে row by row subtract করো।

05 Deadlock — Conditions + Prevention

Deadlock = 2+ processes যেখানে প্রতিটা অন্যের resource-এর জন্য অপেক্ষা করছে এবং কেউ এগোতে পারছে না।

4 Necessary Conditions for Deadlock — সবগুলো একসাথে থাকলে Deadlock

Condition	মানে	উদাহরণ
Mutual Exclusion	একটা resource এক সময়ে একটাই process use করতে পারবে	Printer একসাথে দুজনে use করা যায় না
Hold and Wait	Process একটা resource hold করে অন্যটার জন্য অপেক্ষা করছে	P1 printer নিয়ে scanner-এর জন্য wait করছে
No Preemption	OS জোর করে resource নিতে পারবে না	Process নিজে না ছাড়লে resource যাবে না
Circular Wait	$P1 \rightarrow P2 \rightarrow P3 \rightarrow P1$ — circular dependency	P1 needs P2's resource, P2 needs P3's, P3 needs P1's

Deadlock Handling Methods

Method	Approach	Example
Prevention	৪টা condition-এর একটা নষ্ট করো	No Hold and Wait: সব resource একসাথে চাও নয়তো কিছুই না
Avoidance	Banker's Algorithm — safe state check	Resource দেওয়ার আগে safety check করো
Detection	Deadlock হতে দাও, পরে detect করো	Resource Allocation Graph cycle check
Recovery	Deadlock detect করলে process kill বা resource preempt	Victim selection — roll back process

Starvation কী?	Deadlock থেকে পার্থক্য
Low-priority process indefinitely wait করতে থাকে (never gets CPU)। Deadlock নয় — system চলছে কিন্তু ওই process কখনো execute হচ্ছে না।	Deadlock: multiple processes completely blocked। Starvation: system চলছে, কিন্তু কিছু process always waiting।

BB NOTE: BB MCQ: "Which scheduling causes starvation?" = SJF (long jobs never get CPU). "Banker's Algorithm is used for?" = Deadlock Avoidance। এই দুটো প্রতি বছর আসে।

06 Memory Management — Paging + Virtual Memory

Paging vs Segmentation

Feature	Paging	Segmentation
Division basis	Fixed-size pages (e.g., 4KB)	Variable-size segments (code, data, stack)
External fragmentation	None (fixed size)	Yes (variable sizes leave gaps)
Internal fragmentation	Yes (last page may be partially full)	None
User's view	Physical address based	Logical address based (programmer sees segments)
Address translation	Page table	Segment table

Virtual Memory + Page Fault

- **Virtual Memory:** RAM-এর চেয়ে বড় programs run করার technique। Disk-কে extended RAM হিসেবে ব্যবহার করে।
- **Page Fault:** যখন program একটা page access করতে চায় কিন্তু সেটা RAM-এ নেই (disk-এ আছে)। OS তখন disk থেকে RAM-এ load করে।
- **TLB (Translation Lookaside Buffer):** Page table-এর fast cache। Frequently used page translations store করে। TLB hit = fast, TLB miss = slow (page table access).
- **Thrashing:** System বেশিরভাগ সময় paging করছে, কম সময় actual execution — performance severely drops।

SOLVED —

BB MCQ:

Page size 4KB, 32-bit virtual address, TLB 128 entries 4-way set associative. TLB tag minimum size?

Page size = 4KB = 2^{12} → page offset = 12 bits

Virtual address bits = 32
Page number bits = $32 - 12 = 20$ bits

TLB sets = 128 entries ÷ 4-way = 32 sets = 2^5 → set index = 5 bits

TLB tag = Page number bits – Set index bits = $20 - 5 = 15$ bits

Answer: 15 bits ← BB AME 2023 exact answer

Page Replacement Algorithms

Algorithm	Policy	Fault Rate	Practical?
FIFO	সবচেয়ে আগে আসা page replace করো	High	Simple but poor
Optimal (Belady's)	ভবিষ্যতে সবচেয়ে দেরিতে use হবে এমন page replace	Minimum (optimal)	NOT practical — future unknown
LRU	সবচেয়ে আগে last used page replace করো	Near-optimal	Most commonly used in practice
Clock (Second Chance)	Reference bit check করো before replacing	Moderate	Good compromise — used in Linux

BB NOTE: BB MCQ: "Page replacement with minimum page faults?" = Optimal (Belady's) — but impractical. Practical-এ LRU। "Page fault occurs when?" = Requested page is NOT in memory।

07 Semaphore + Critical Section (Other Bank Written)

Critical Section Problem

Multiple processes যখন shared data access করে — inconsistency হতে পারে। Critical Section = shared resource access করার code segment।

- ৩টা Requirements:

- Mutual Exclusion: একসময়ে একটাই process critical section-এ থাকবে
- Progress: কেউ critical section-এ না থাকলে যে যেতে চাইছে সে যাবে
- Bounded Waiting: একটা process infinitely wait করবে না

Semaphore — Solution

Type	Value	Use case	Operations
Binary Semaphore (Mutex)	0 বা 1	Mutual exclusion — critical section protect	wait(S): S-- if S<0 block। signal(S): S++ if blocked process → wake it
Counting Semaphore	0 থেকে N	Multiple instances of resource manage	Same wait/signal — N instances track করে

SOLVED — Producer-Consumer with Semaphore Semaphores needed: mutex = 1 (mutual exclusion for buffer) empty = n (number of empty slots) full = 0 (number of full slots)
 Producer: wait(mutex) [add item] signal(mutex)
 Consumer: wait(empty) [remove item] signal(empty)
 wait(full) signal(full)

Semaphore	Mutex	Monitor
Any integer value. More general.	Binary (0/1). Specifically for mutual exclusion.	High-level synchronization construct. Language supported.
wait() and signal() operations	lock() and unlock() operations	Procedures are automatically mutually exclusive

08 BB MCQ Quick-Fire Sheet

Question	Answer	Source
Process needs I/O → goes to which state?	Waiting (Blocked)	BB AD 2025 MCQ
Banker's Algorithm is used for?	Deadlock Avoidance	BB MCQ Every Year
Which scheduling causes starvation?	SJF (long processes never get CPU)	BB MCQ 2019
Best scheduling for time-sharing?	Round Robin	BB MCQ Every Year
Multiprogramming advantage?	Execute more jobs in same time period	BB MCQ 2021
Page fault occurs when?	Requested page is NOT in memory	BB MCQ 2023
4 Deadlock conditions?	Mutual Exclusion + Hold&Wait + No Preemption + Circular Wait	BB MCQ Every Year
Need = ?	Max – Allocation	Banker's Algorithm formula
TAT = ?	CT – AT (Completion Time – Arrival Time)	Scheduling formula
WT = ?	TAT – BT (Turnaround – Burst Time)	Scheduling formula
OS runs in which mode?	Kernel mode (supervisor/privileged mode)	BB MCQ 2022
Thrashing occurs when?	System spends more time paging than executing	BB MCQ 2023
TLB is used for?	Fast address translation (cache for page table)	BB MCQ 2023
Context switch is overhead — which algo worst?	Round Robin (most frequent switching)	BB MCQ 2024
Semaphore wait() = ?	Decrement; if negative, block the process	Critical Section question
Safe state means?	There exists a safe sequence where all processes can complete	Banker's concept
SRTF (Shortest Remaining Time First) is preemptive version of?	SJF (Shortest Job First)	BB MCQ advanced
CPU scheduling in OS manages?	Process queue — decides which process gets CPU next	BB MCQ basic
Virtual memory is combination of?	RAM + Disk (secondary storage)	BB MCQ 2022
LRU page replacement stands for?	Least Recently Used	BB MCQ page replacement

09 Written Question Approach — Type by Type

Type দেখলে...	প্রথমেই করো	Format
"CPU Scheduling", "Gantt chart", "FCFS/SJF/RR", "TAT", "WT"	Write: AT BT CT TAT WT column table + Formulas	Table → Gantt Chart → Calculate CT for each → $TAT=CT-AT$ → $WT=TAT-BT$ → Average
"Banker's Algorithm", "safe state", "safe sequence", "Need matrix"	Write: Need = Max – Allocation. Calculate Need matrix first.	Need matrix → Available → Safety Algorithm step-by-step → Safe sequence
"Deadlock", "conditions", "prevention"	Write: 4 conditions: ME + H&W + NP + CW	Define each + example → Handling methods (prevention/avoidance/detection/recovery)
"Semaphore", "critical section", "mutex"	Write: 3 requirements: ME + Progress + Bounded Waiting	Define → Semaphore types → wait()/signal() → example (producer-consumer)
"Virtual memory", "paging", "page fault"	Write: Definition → How it works → Page fault process	Definition → Paging vs Segmentation → Page replacement (FIFO/LRU/Optimal)

TIP: Scheduling question-এ সবসময় Gantt Chart দিয়ে শুরু করো। Calculation ভুল হলেও Gantt Chart-এর জন্য marks পাবে। Banker's-এ Need matrix এবং Safety Algorithm-এর প্রতিটা step clearly দেখাও — examiner step count করেন।

10-Day Practice Schedule

দিন	Focus	কী করবে	লক্ষ্য
Day 1	Scheduling Formulas	$TAT=CT-AT$, $WT=TAT-BT$ — এই formulas দিয়ে 3টা example করো	Formula মুখস্থ
Day 2	FCFS + SJF	FCFS এবং SJF দুটো same data দিয়ে করো — পার্থক্য দেখো	Gantt Chart by hand
Day 3	Priority + Round Robin	BB AP 2023 exact problem solve করো, তারপর RR করো	Both algorithms
Day 4	Scheduling Mixed	Arrival time সহ 4 processes — FCFS + SJF দুটোই	Speed 15 min/problem
Day 5	Banker's Basics	Need matrix calculation 5 problems	Need formula instant
Day 6	Banker's Safety Algorithm	BB AD 2025 Q4 exact solve করো without notes	Safety sequence
Day 7	Deadlock	4 conditions মুখস্থ + Banker's vs Detection কখন	Concept clear
Day 8	Memory Management	Page fault, TLB calculation, Paging vs Segmentation	MCQ ready
Day 9	All BB OS Questions	Section 15 থেকে সব OS written questions — without notes	Exam simulation
Day 10	Timed Mock	Scheduling (15 min) + Banker's (15 min) — ঘড়ি ধরে	Exam timing

10 All Previous Year OS Questions (2019–2026)

Total: 21 questions | MCQ: 12 | Written: 9

BB NOTE: BB AD questions সবচেয়ে বেশি priority | Written questions কাগজে নিজে solve করো — শুধু পড়লে হবে না।

Bangladesh Bank — OS Questions

Year	Exam	Type	Question
2025	BB AD (DU)	Written	Banker's Algorithm: 5 processes P0-P4, 3 resource types A(10), B(5), C(7). Find Need matrix. Is system in safe state?
2025	BB AD (DU)	MCQ	A process needs I/O operations, it switches to which state? Ans: Waiting (Blocked)
2025	BB AP (BIBM)	MCQ	Priority Scheduling (Non-Preemptive): P1(BT=10,P=3), P2(BT=1,P=1), P3(BT=2,P=4), P4(BT=1,P=5), P5(BT=5,P=2). Average TAT? Ans: 12
2023	BB AP (BIBM)	MCQ	Page fault occurs when: Ans: Requested page is NOT in memory
2023	BB AME (BIBM)	MCQ	TLB is 4-way set associative with 128 entries. Page size 4KB. Virtual address 32-bit. TLB tag minimum size? Ans: 15 bits
2021	BB AP (DU)	MCQ	Multiprogramming systems: execute more jobs in same time — which is correct? Ans: Execute more jobs in the same time
2019	BB AP (DU)	Written	Explain different CPU scheduling algorithms with examples. Compare FCFS and SJF.
2019	BB AP (DU)	MCQ	Which scheduling algorithm causes starvation? Ans: SJF (short jobs always prioritized, long jobs wait forever)

Other Banks — CPU Scheduling Questions

Year	Exam	Type	Question
2021	RAKUB PO	Written	Banker's Algorithm: 5 processes, 3 resource types. Find Need matrix and determine safe state and safe sequence.
2020	Combined SB/JB	Written	CPU Scheduling: FCFS/SJF — calculate TAT, WT for given process table. Draw Gantt chart.
2020	ANE JBL	MCQ	Process goes to Blocked/Waiting state when? Ans: When it needs I/O operation
2022	Combined 3 Bank	Written	Explain virtual memory. What is page fault? Difference between paging and segmentation.
2024	Combined 3 Bank	Written	CPU Scheduling with arrival time: P1(AT=0,BT=7), P2(AT=2,BT=4), P3(AT=4,BT=1), P4(AT=5,BT=4). FCFS and SJF Gantt chart and TAT/WT.
2026	Sonali/Janata	MCQ	Context switch overhead — which scheduling algorithm has highest overhead? Ans: Round Robin (frequent switching)
2026	BSCS Sonali	Written	Compare FCFS, SJF, Round Robin scheduling with example. Which is best for time-sharing systems?

Other Banks — Deadlock + Memory + Semaphore

Year	Exam	Type	Question
2022	ANE JBL	Written	Explain Deadlock: necessary conditions, detection, and prevention methods.
2022	ADA SBL	MCQ	What is the Banker's algorithm used for? Ans: Deadlock avoidance

Year	Exam	Type	Question
2022	Combined 3 Bank	MCQ	Page replacement algorithm with minimum page faults? Ans: Optimal (Belady's) — impractical; LRU practical
2024	Combined 5 Bank	MCQ	Critical section problem solution must satisfy: Mutual exclusion, Progress, Bounded waiting — all 3?
2023	RAKUB BIBM	Written	Explain semaphore and mutex. How do they solve critical section problem?
2023	Rupali Bank	MCQ	Thrashing in operating system occurs when? Ans: System spends more time paging than executing processes

OS Master Strategy:

CPU Scheduling = Gantt Chart হাতে আঁকো, TAT/WT calculate করো। Rote নয়, practice।
 Banker's Algorithm = Need matrix first, তারপর Safety Algorithm step-by-step। Work track করো।
 Deadlock = 4 conditions সবসময় মুখস্থ। Banker's = Avoidance। Detection ≠ Avoidance।
 MCQ = Process States transition + Page Fault + TLB — এই তিনটা প্রতি বছর।